Installation and Operation Manual

GreenPower

GPWF1-40.5kV

SF6 Outdoor Load Break Switch





About Us»

GreenPower is a manufacturer of medium- and low-voltage switchgear and components, with in-house engineering, manufacturing and testing capabilities. Our manufacturing entity, Zhejiang GreenPower Electric Co., Ltd., was established in 1992. International business and export operations are handled by Zhejiang GreenPower I&E Co., Ltd.

Our product portfolio covers medium-voltage switchgear from 12 kV to 40.5 kV, including air-insulated metal-clad switchgear (AIS), cubicle-type gas-insulated switchgear (C-GIS), ring main units (RMU), vacuum circuit breakers, and a broad range of switchgear components and accessories. We also manufacture low-voltage withdrawable switchgear systems with ratings up to 6300 A.

Engineering and manufacturing are central to what we do. Our production and testing capabilities include automated sheet-metal processing, CNC laser cutting, 3D five-axis laser welding for stainless-steel gas tanks, automatic helium leak detection, and a fully shielded partial-discharge test room. These capabilities support consistent manufacturing quality from enclosure fabrication through assembly and final testing.

For customers moving toward lower-impact power distribution systems, GreenPower offers SF6-free solutions including dry-air insulated RMUs and nitrogen-insulated C-GIS, alongside conventional gas-insulated solutions where required. These technologies help customers respond to increasingly stringent environmental requirements for medium-voltage switchgear.

We also support project-specific engineering based on customer drawings and specifications, including non-standard dimensions, special voltage and frequency requirements, high-altitude applications, customized component selection and other application-specific configurations.

Our goal is simple: to be the switchgear manufacturer our customers return to — because the equipment is properly engineered, reliably manufactured and built for long-term service.

01 Product Overview

The GPWF1-40.5 outdoor SF_6 load break switch is intended for distribution networks with a maximum system voltage of 40.5kV, rated current of 630A and rated frequency of 50/60 Hz. It is capable of making and breaking load current and overload current in power systems. When equipped with an intelligent controller, the load break switch can be used in automated distribution systems to realize automatic sectionalizing and fault isolation.

02 Service and Environmental Conditions

- Altitude: ≤ 3000 m.
- Ambient air temperature: -40℃ to +50℃ , maximum daily temperature variation ≤ 25℃ .
- Relative humidity: daily average ≤ 95 %, monthly average ≤ 90 %.
- Maximum wind speed: ≤ 34 m/s.
- Seismic intensity: ≤ 8 degrees.
- Icing thickness: < 10 mm.
- Snow thickness: < 35 mm.
- Solar irradiance: < 0.1 W/cm^2.
- Pollution level: Class III or Class IV.

Note: The installation site shall be free from fire and explosion hazards, serious chemical corrosion, and frequent severe vibration.

03 Model Code and Explanation

GP WF 1 - 40.5 / 630 - □

- Rated short-circuit withstand current (kA)
- Rated current (A)
- Rated voltage (kV)
- Design No.
- Outdoor SF_6 load break switch
- Company feature code

04 Main Technical Parameters

Item			Unit	Data
Rated voltage			kV	40.5
Rated current			A	630
Rated frequency			Hz	50 / 60
Rated insulation level (at SF_6 gas pressure 0.07 MPa / 20℃)	Power frequency withstand voltage (1 min)	Phase-to-phase / earth	kV	95
		Across open contacts	kV	110
	Lightning impulse withstand voltage	Phase-to-phase / earth	kV	185
		Across open contacts	kV	215
	Rain withstand voltage test		kV	70
Rated load breaking current (at 0.07 MPa / 20℃)			A	630
Rated load breaking operations			times	≥ 400
Load breaking current at zero gauge pressure			A	630
Rated short-time withstand current			kA/4s	20
Short-time withstand current (peak)			kA	50
Rated making current (peak)			kA	50
Rated making operations			times	3
Rated closed-loop breaking current			A	630

Item		Unit	Data
Rated active load breaking current		A	630
Rated cable charging breaking current		A	25
Rated line charging breaking current		A	16
Rated no-load transformer breaking current		A	21
Rated operating gas pressure (with low-pressure interlock)		MPa	0.07 (0.15)
Main circuit resistance per phase		μΩ	≤ 150
Annual relative gas leakage rate		% / year	≤ 0.5
SF_6 Gas moisture content	Factory delivery value	ppm	≤ 150
	In-service value	ppm	≤ 300
Mechanical endurance		times	6000
Rated operating voltage and auxiliary circuit rated voltage		V	DC 24

05 Insulation Gas

The load break switch uses high-purity SF_6 gas, specially processed for electrical equipment, as both the arc-quenching and insulation medium. During the whole service life of the switch, the moisture content inside the sealed tank is controlled to be less than 300 ppm.

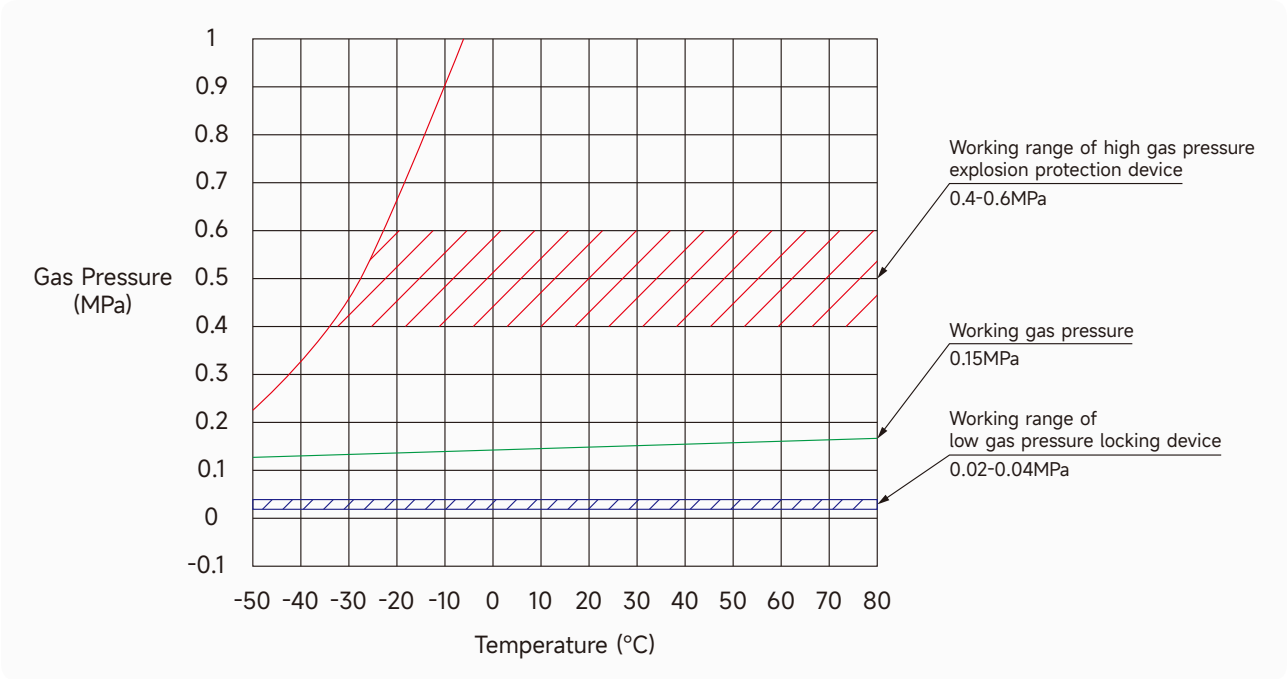
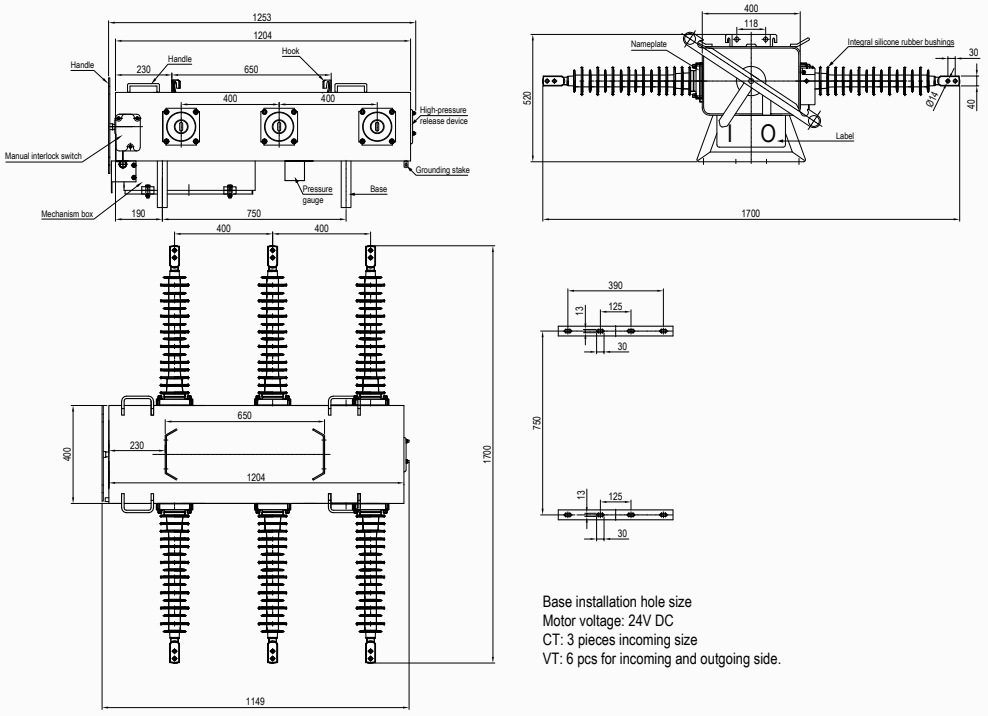


Figure 1 SF_6 gas temperature-pressure curve

- SF_6 gas filling pressure at 20℃ : 0.15 MPa.
- Rated operating pressure: 0.15 MPa.
- Operating pressure of high-pressure relief and explosion-venting device: 0.4-0.6 MPa.
- Operating pressure of low-pressure lockout and alarm device: 0.02-0.04 MPa.

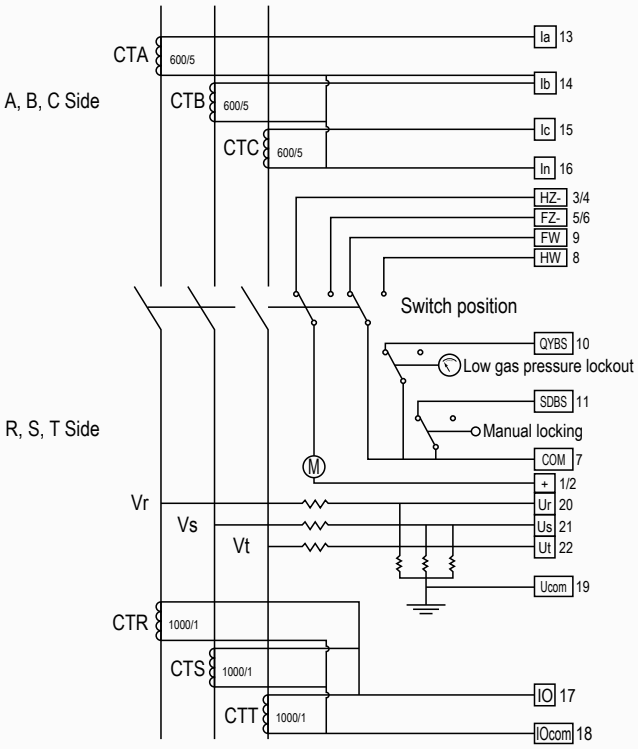
06 Structural Features



Insulating bushing creepage distance 1400mm

Base installation hole size
Motor voltage: 24V DC
CT: 3 pieces incoming size
VT: 6 pcs for incoming and outgoing side.

07 Electrical Schematic and Wiring Diagram



08 Storage

Before installation and use, the product shall be stored and preserved in accordance with the following requirements:

- **Short-term Storage (≤ 3 months)**
  a) Do not open the package.
  b) Store the product away from direct sunlight and high-temperature environments.
  c) Do not store the product in locations with high humidity or standing water.
- **Long-term Storage (> 3 months)**
  a) Store the product in a cool, dry and well-ventilated place, protected from direct sunlight.
  b) Avoid locations with high humidity or standing water.
  c) During long-term storage, the battery in the control box will self-discharge and its performance will decrease; it should be inspected and recharged periodically.

09 Installation

Before installation, the following items shall be checked:

- After unpacking, verify the contents and accessories against the packing list.
- Visually inspect the switch body to ensure that the silicone rubber bushings, open/close position indicators, manual operating handle and gas pressure gauge are intact and undamaged.
- Using the gas pressure gauge, check that the SF_6 pressure is within the specified range (rated pressure ± 0.01 MPa).
- Perform at least five manual close/open operations to confirm that the mechanism operates smoothly without jamming.
- Connect the switch tank to the control box with the control cable and power cable, energize the control power supply, and verify that all indicating lamps operate correctly.
- If no abnormal condition is found in the above checks, the product can be installed on site.
- From the control box, perform at least five motor-operated close/open cycles and confirm that the switch operates normally.

Caution: After unpacking, do not lift or move the switch by the silicone rubber bushings or by the operating handle, in order to avoid damage or malfunction.

10 Inspection During Operation

After the switch is installed and put into service, the following routine or periodic checks shall be carried out:

- Visually check whether the switch tank, bushings and accessories are in normal condition and free from visible damage or abnormalities.
- Before each closing or opening operation, make sure that the SF_6 gas pressure indicated on the gauge is within the normal range.
- Check whether the open/close status of the switch body is consistent with the indication on the control box lamps.
- Confirm that all indication lamps and switches on the control box are functioning correctly.
- Use the battery test button to check whether the battery status indicator and the charging indicator lamp are normal.
- When necessary, use a multimeter to measure the battery voltage and charging current to assess the battery condition.

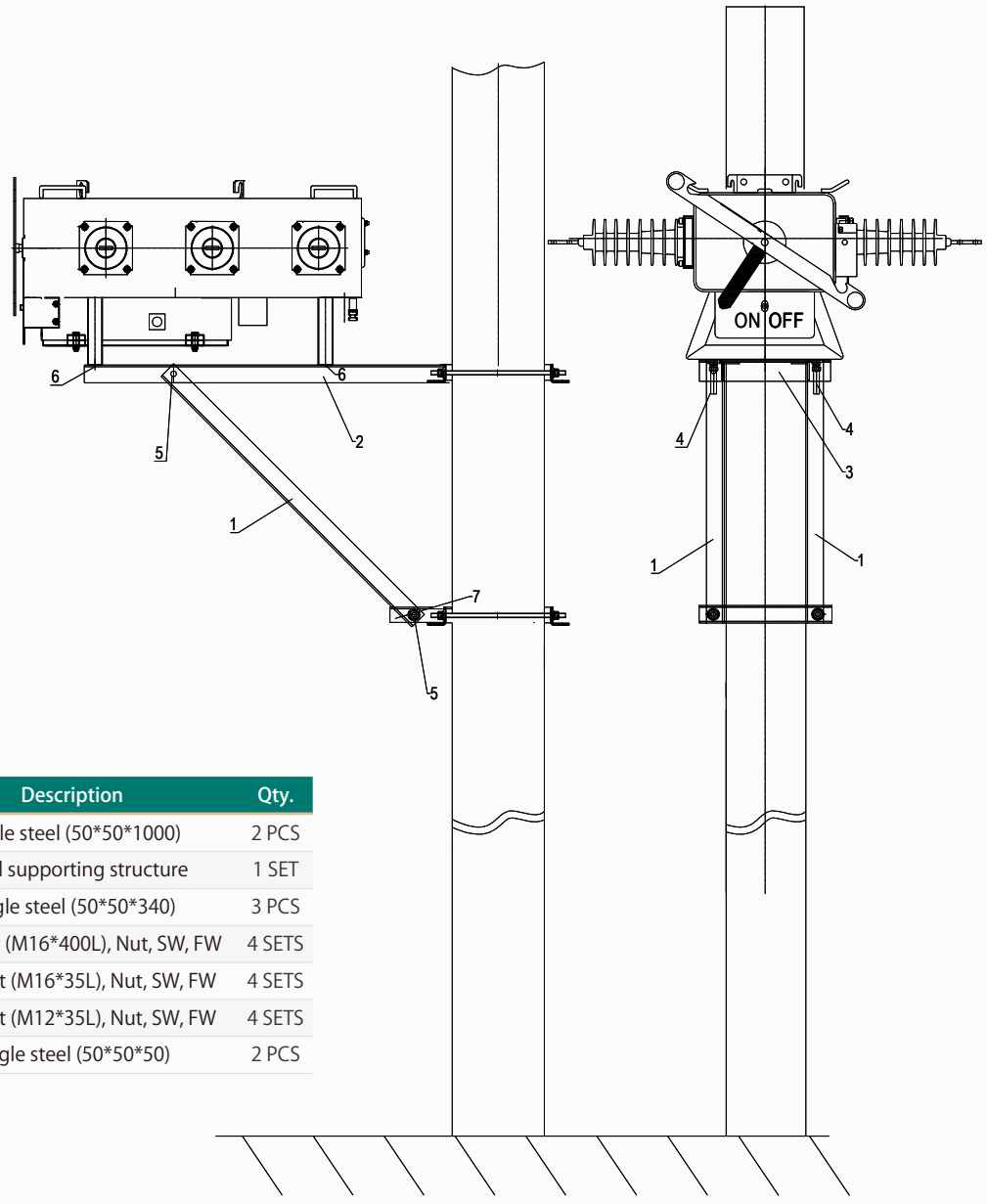
11 Maintenance

The switch is completely assembled, adjusted and tested before leaving the factory, and under normal operating conditions it does not require routine maintenance. Maintenance is required only in the following cases:

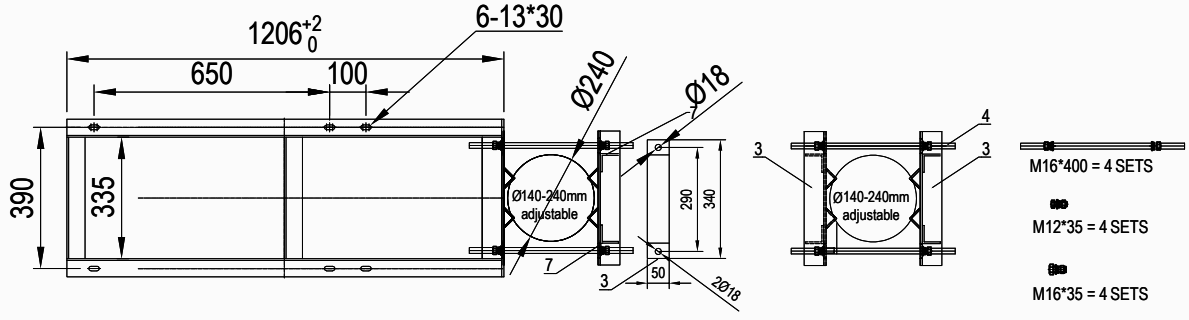
- After the switch has been in service for some time, clean the bushing surfaces periodically to remove dust and pollution so as to maintain good insulation performance.
- If any defects caused by product quality or SF_6 leakage occur within five years from the date of shipment, the manufacturer will be responsible for repair or replacement.

For reference to SF_6 gas conditions during service, see the SF_6 gas temperature-pressure curve shown in Figure 1.

12 Installation Diagram



No.	Description	Qty.
1	Angle steel (50*50*1000)	2 PCS
2	Fixed supporting structure	1 SET
3	Angle steel (50*50*340)	3 PCS
4	M16 Bolt (M16*400L), Nut, SW, FW	4 SETS
5	M16 Bolt (M16*35L), Nut, SW, FW	4 SETS
6	M12 Bolt (M12*35L), Nut, SW, FW	4 SETS
7	Angle steel (50*50*50)	2 PCS



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Zhejiang Greenpower Electric Co., Ltd
Zhejiang GreenPower I&E Co., Ltd
Professional Power Equipment Provider

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GreenPower

Contact us

www.sinogp.com
gp@sinogp.com
No.827 Wenzhou Avenue, Wenzhou, Zhejiang, 325011, P. R. China
0577 8550 0968
+86 189 8978 2765